

RAD-SMART (Radiotherapy AI assistive Dynamic Scheduling, Machine Allocation and Resource Tracking)

1. Background and the Problem

Radiotherapy is different from a conventional outpatient appointment system. A patient undergoing radiotherapy typically attends the department every working day (Monday to Friday) for several days or weeks (typically 5-7 weeks for curative cases and 1-2 weeks for palliative cases). A single treatment machine may therefore treat a large number of existing patients each day (typically 70-90 patients per day) while simultaneously accommodating new patients starting treatment, urgent palliative cases and complex procedures.

At present, patient appointments in our department are largely allocated manually by the radiation therapy technologists. Appointment times are not based on a systematic assessment of the actual time required for each treatment or the predicted workload of the machine. As a result, too many patients may be asked to report during the same period, while other periods are relatively underutilized. Patients consequently experience long and unpredictable waiting times despite having been given an appointment.

The scheduling problem is more complex than simply distributing patients evenly throughout the working day because **different patients consume different amounts of machine time and have different scheduling constraints.**

For example, a routine treatment may require only a few minutes (typically 7-15 minutes), whereas a new treatment start or a complex procedure such as stereotactic body radiotherapy (SBRT), stereotactic radiotherapy (SRT), total body irradiation (TBI) or craniospinal irradiation (CSI) may require considerably more time (20-60 minutes). Complex treatments also require availability of senior clinical staff and therefore need to be performed during defined working hours (10AM to 5PM).

Certain patients also require prioritization because of clinical or logistical circumstances. These include palliative patients, elderly patients, inpatients, patients staying in institutional accommodation or respite facilities, and patients dependent on public transport for returning home.

Another important operational decision is **how many new patients should be started on a particular day.** Every new patient started today adds to the machine workload for subsequent days or weeks. At present, this decision is difficult to make objectively. Ideally, the number of new starts should be decided approximately two days in advance based on the current machine workload, treatment durations and the number of patients expected to complete treatment over the coming days.

The department also has resources such as breast boards and Active Breathing Coordinator (ABC) equipment that may be shared between the CT simulator and treatment machines. A patient may therefore be ready for treatment but unable to proceed because the required accessory is being used elsewhere.

The complexity of this problem will increase further as the department expands from the current single-machine workflow to multiple treatment machines with different capabilities.

The fundamental problem is therefore not appointment booking. It is dynamic allocation of patients, machine time, staff availability and shared resources under multiple clinical and operational constraints.

2. What We Require

We require a system capable of converting the day's radiotherapy workload into an **executable patient-level schedule**.

Instead of simply recording an appointment entered manually by a technologist, the system should determine an appropriate treatment time based on the predicted duration of the patient's treatment and all applicable operational constraints.

The system should consider:

- current machine occupancy;
- expected duration of treatment for individual patients;
- number of remaining treatment fractions;
- anticipated treatment-completion dates;
- new treatment starts to be completed before 5PM;
- patient-priority categories (Paying patients/ elderly/ inpatient etc.);
- urgent/palliative treatments which needs starting treatment same day as consultation;
- patient-specific logistical constraints (eg: dependent on public transport for commute);
- protected time periods for complex procedures to ensure availability of appropriate staff supervision;
- availability of shared accessories;
- Daily slot for blood irradiation (1:30 to 2PM)
- machine downtime and operational delays; and
- availability and workload of multiple treatment machines.

Some scheduling rules are **hard constraints**. For example, complex treatments such as SBRT, SRT, TBI and CSI may need to be performed during a protected period when appropriate senior staff are available (10AAM to 5PM). Blood irradiation has a dedicated time slot. New treatment starts should be completed within the defined working period (before 5PM).

Other requirements represent **soft constraints**. For example, an elderly patient or someone dependent on public transport may preferentially receive an earlier appointment if capacity permits.

These rules should be configurable rather than permanently hard-coded so that the same platform can eventually be adapted to other radiotherapy departments.

3. Proposed Solution

RAD-SMART: Radiotherapy AI-Assisted Dynamic Scheduling, Machine Allocation and Resource Tracking.

RAD-SMART will function as an intelligent operational layer between the patient list and the treatment machines.

The system will receive information about the patients who need treatment, their treatment characteristics, machine availability and operational constraints and generate an optimized schedule for the day.

Conceptually:

Patient and treatment data

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Machine + staff + accessory availability

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Institutional scheduling rules and patient priorities

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RAD-SMART Scheduling Engine

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Executable patient-wise schedule

For example, instead of 15 patients being arbitrarily told to arrive at 9 AM, the system might assign:

Patient A	—	09:00
Patient B	—	09:15
Patient C (new start)	—	09:30
Patient D	—	09:50

The objective is not merely to fill every minute of machine time. The optimization should balance several goals simultaneously:

minimize patient waiting time + maximize machine utilization + accommodate priority patients + satisfy procedural constraints + avoid resource conflicts + minimize staff overtime.

4. Predicting Treatment Duration

An important component of RAD-SMART will be predicting how much machine time an individual patient is likely to require.

Treatment time varies according to factors such as treatment technique, treatment site, whether the patient is undergoing the first fraction, imaging requirements and complexity of treatment.

Initially, the system can use predefined average treatment durations. As historical operational data become available, a predictive model can learn the actual treatment duration associated with different types of treatments.

The scheduling engine would therefore allocate **machine minutes rather than simply counting patients**.

This distinction becomes particularly important when multiple machines are available. Forty simple treatments on one machine may represent a smaller workload than thirty complex treatments on another.

5. Forecasting Machine Occupancy and New Starts

RAD-SMART should look beyond today's schedule.

For every patient currently undergoing radiotherapy, the system knows the number of remaining fractions and expected treatment-completion date. It can therefore estimate how machine occupancy is likely to change over the coming days.

For example:

Monday:	High	occupancy	—	recommend	2	new	starts
Tuesday:	High	occupancy	—	recommend	3	new	starts
Wednesday:	Several patients completing treatment			—	recommend	6	new starts
Thursday:	Moderate occupancy — recommend 5 new starts						

The system can therefore provide a recommendation approximately two days in advance regarding how many new patients can be accommodated without creating excessive congestion.

This transforms new-start allocation from a subjective decision into a prospective capacity-planning process.

6. Dynamic Rescheduling

Radiotherapy departments are inherently unpredictable.

A machine may develop a technical fault. A patient may not attend. An urgent palliative patient may need to be accommodated. A complex treatment may take longer than anticipated.

A conventional appointment system cannot respond intelligently to these changes.

RAD-SMART should continuously compare the planned schedule with actual progress.

If the machine develops a 30-minute delay, for example, the system should recalculate the remaining schedule while preserving important constraints and priorities.

The revised expected treatment times could then be displayed to staff and, in future versions, communicated to patients.

Thus, RAD-SMART evolves from a **static scheduler into a real-time radiotherapy operations management system.**

7. Shared Resource Tracking

The scheduling engine should also account for resources required for treatment.

Examples include:

- breast boards;
- ABC equipment;
- specialized immobilization devices; and
- other accessories shared between the simulator and treatment machines.

If an accessory is scheduled for use simultaneously on the simulator and treatment machine, RAD-SMART should identify the conflict before it occurs and either modify the schedule or alert the treatment team.

8. Expansion to Multiple Treatment Machines

The platform should be designed from the outset for more than one machine.

When multiple machines are available, RAD-SMART should determine both:

Which machine should treat the patient?

and

At what time should the patient be treated?

Allocation can consider machine capabilities, treatment technique, expected treatment duration, existing workload, required accessories and machine availability.

The objective would be to balance actual predicted workload across machines rather than simply allocating an equal number of patients to each machine.

9. Expected Impact

For Patients

The primary benefit is **predictability**. Patients should be able to arrive close to their actual treatment time rather than spending prolonged periods waiting outside the treatment machine.

This is particularly important because radiotherapy patients attend repeatedly, often for several weeks. Even a modest reduction in daily waiting time can translate into many hours saved over an entire treatment course.

For Radiation Therapy Technologists

RAD-SMART reduces the burden of manually deciding appointment times, accommodating new starts and continuously reorganizing the patient queue when unexpected events occur.

For Clinicians

Complex procedures can be reliably scheduled during periods when appropriate clinical supervision is available, reducing the need for ad-hoc coordination.

For the Department

The system should improve machine utilization, reduce periods of unnecessary congestion or idle time, reduce overtime and provide objective information for future capacity planning.

10. How Success Will Be Measured

The impact of RAD-SMART can be evaluated objectively by comparing departmental workflow before and after implementation.

Potential metrics include:

- median patient waiting time;
- 90th percentile waiting time;
- percentage of patients treated within ± 15 or ± 30 minutes of their assigned appointment;
- percentage of new starts completed within the designated time;
- daily overtime;
- number of scheduling conflicts;
- number of accessory-related delays;

- treatment-time prediction accuracy; and
- discrepancy between predicted and actual daily machine occupancy.

The ultimate goal is straightforward:

The right patient should reach the right radiotherapy machine, with the required resources and staff available, at a predictable time—while the department uses its limited treatment capacity as efficiently as possible.

RAD-SMART aims to convert radiotherapy scheduling from a largely manual and reactive process into a **predictive, constraint-aware and dynamically optimized operational workflow**.

11. Summary

RAD-SMART will generate executable patient treatment schedules by integrating:

- existing machine occupancy
- predicted treatment duration
- anticipated treatment completion dates
- patient-priority categories (paying patients/ elderly/ MHRC/ IP/ dormitory)
- new treatment starts before 5PM
- protected procedural time slots (12noon to 1:30pm) for complex treatments (SBRT /SRT /TBI /CSI) requiring staff supervision
- Dedicated slot for blood irradiation (1:30 to 2PM)
- machine capabilities (Versa HD/ Halcyon)
- shared treatment and simulation accessories (breast board/ ABC)
- Real-time recovery based on operational delays or machine downtime.